Why Governments Do Not Build Funds: Political Turnover, Fund Raiding and the Missing Asset Path

Progress report

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Abstract

In the planner benchmark of Battaglini and Coate (2008), a government accumulates assets until the interest finances spending and taxes can be zero forever. Real governments, Indian states among them, almost never do this. I study one friction: after a change of government, a successor can divert part of any accumulated fund. Over six model versions this gives (i) a turnover threshold for saving and three fiscal regimes (borrow, balanced budget, save); (ii) a precautionary version in which the turnover risk a state can tolerate scales with the *variance* of spending shocks while the cost of raiding is proportional to the raided share; (iii) a fungibility result, in which legal protection of a sinking fund matters only when borrowing headroom is small; and (iv) the latest and main result: in an infinite-horizon economy with spending shocks, *any* positive expected raiding removes the planner's long-run anchor. There is no steady state with a finite average marginal cost of public funds. With a debt ceiling, a steady state exists and the ceiling must bind with positive probability, governed by an exact "wedge accounting" identity.

1 Question and motivation

Benchmark. With $\beta(1+r)=1$ and spending uncertainty, the planner in Battaglini and Coate (2008) (see also Aiyagari, Marcet, Sargent and Seppälä 2002) front-loads taxes and builds up assets until the interest covers spending; taxes are then zero forever.

Puzzle. Most governments carry debt (Indian general government debt is about 80% of GDP). Large public funds are rare: Norway, Gulf states, Chile.

Question. Why don't governments accumulate assets as the benchmark prescribes, and which institutions allow some to do so?

Mechanism. The incumbent stays in office with probability q. After turnover, the successor diverts a share φ of any accumulated assets (φ measures how well the fund is protected). The incumbent's effective return on saving is $(1+r)\,[\,q+(1-q)(1-\varphi)\,]$.

Setting: Indian states. Indian states combine high debt, small Consolidated Sinking Funds (CSF) at the RBI, strong anti-incumbency, a return to the Old Pension Scheme in several states (liabilities pushed onto successors), and a net borrowing ceiling set by the Centre.

Closest literature.

- *Battaglini and Coate (2008)*, the main overlap. Pork for minimum winning coalitions keeps debt above the planner's level. Their friction is *within-period*: part of extra revenue goes to pork whoever governs. Mine is *between-period*: the stock of assets is lost only on turnover. A government sure to stay in office saves at the planner's level here.

- *Alesina and Tabellini (1990)*: alternating parties that disagree over spending over-borrow. I use their disagreement channel in Version 2.
- *Battaglini (2014)*: dynamic probabilistic voting with debt as the state; a template for making q endogenous.
- *Battaglini (2026)*: alternation with a durable state can produce cycles and chaos in Markov equilibrium. This is relevant to my Version 3 numerics.

2 Common environment

Households have quasi-linear preferences $u = c + A\,v(g) - h(l)$. With a linear labour tax τ and wage 1, labour supply is $l(\tau) = (1-\tau)^{\varepsilon}$, revenue is $R(\tau) = \tau(1-\tau)^{\varepsilon}$ (Laffer peak at $\tau_p = 1/(1+\varepsilon)$), and household surplus is $U(\tau) = (1-\tau)^{1+\varepsilon}/(1+\varepsilon)$. The marginal cost of public funds (MCPF) is

$$\lambda(\tau) = \frac{-U'(\tau)}{R'(\tau)} = \frac{1-\tau}{1-(1+\varepsilon)\tau},$$

which rises with τ; this is the reason to smooth taxes. The budget is $g_t + (1+r)b_t = R(\tau_t) + b_{t+1}$, with $b < 0$ meaning assets. After turnover, assets are cut by the raid map

$$T(b) = \begin{cases} (1-\varphi)\,b & b < 0 \\ b & b \geq 0 \end{cases}$$

so debt is never raided.

3 Version 1: a two-period threshold

Set-up. Two periods; spending fixed at G; zero initial debt; a benevolent period-1 government saves $a \geq 0$. Budgets: $R(\tau_1) = G + a$; in period 2, $R(\tau_2) = G - (1+r)a$ if the incumbent stays, and $G - (1-\varphi)(1+r)a$ if not.

Derivation. Maximise $U(\tau_1) + \beta[\,qU(\tau_2^S) + (1-q)U(\tau_2^L)\,]$. Using $d\tau/dx = 1/R'(\tau)$, the first-order condition is

$$\lambda(\tau_1) = \beta(1+r)\big[\,q\,\lambda(\tau_2^S) + (1-q)(1-\varphi)\,\lambda(\tau_2^L)\big].$$

At $a = 0$ all tax rates are equal, so λ cancels. The first rupee is saved iff $\beta(1+r)[1-(1-q)\varphi] > 1$, that is,

$$(1-q)\,\varphi < \kappa \equiv 1 - \frac{1}{\beta(1+r)}.$$

Borrowing at $a = 0$ costs $\beta(1+r)\lambda > \lambda$ when $\kappa > 0$, so it never happens.

Result. Expected raiding $(1-q)\varphi$ is compared with the patience gap κ. With $\beta = 0.96$ and $r = 5\%$, $\kappa \approx 0.8\%$, so even small raiding risk stops saving. ε does not enter the threshold. Above the threshold the budget is balanced: Version 1 explains the absence of assets, but not debt.

4 Version 2: a disagreeing successor and three regimes

Set-up. Parties L and R each value only their own public good and both dislike tax distortions; g is chosen with $Av'(g) = \lambda(\tau)$. L governs in period 1. When the period-2 government must

repay one more rupee, it raises a share s through taxes and cuts its own good by $1-s$. From the first-order conditions,

$$s = \Big[1 + \frac{\lambda'(\tau)}{-A\, v''(g)\, R'(\tau)}\Big]^{-1}.$$

Let $d = 1 - s$: the share of a debt burden that lands on spending L does not value (disagreement).

Derivation. At $b = 0$, if L borrows a rupee, under R it bears only the tax part s. So borrowing pays iff $\beta(1+r)[q + (1-q)s] < 1$, i.e. $(1-q)d > \kappa$. If L saves a rupee, under R a share φ is raided first and R uses only s of the rest to cut taxes. So saving pays iff $\beta(1+r)[q + (1-q)(1-\varphi)s] > 1$, i.e. $(1-q)[d + \varphi(1-d)] < \kappa$.

Regime	Condition at $b = 0$	Example
Borrow	$(1-q)\, d > \kappa$	Indian states
Balanced budget	in between	
Save	$(1-q)\,[d + \varphi(1-d)] < \kappa$	Norway

Results. Turnover and disagreement create debt. Raiding blocks assets but cannot create debt on its own, and the balanced-budget band exists only because $\varphi > 0$. With $\kappa = 0$ any $q < 1$ gives debt, the pure Alesina–Tabellini case. With $v = \log$, $d \approx 0.014$, 0.028, 0.058 for $\varepsilon = 0.25$, 0.5, 1. A grid search matched the predicted regime in all 17 cases tried.

5 Version 3: infinite horizon, and why the benchmark had to change

Set-up. The state is debt b. With symmetric parties, V (party in power) and W (party out of power) satisfy

$$V(b) = \max_{\tau, g, b'} U(\tau) + A\, v(g) + \beta\big[q\, V(b') + (1-q)\, W(T(b'))\big],$$

$$W(b) = U(\tau^*(b)) + \beta\big[q\, W(b^*(b)) + (1-q)\, V(T(b^*(b)))\big],$$

and I look for a Markov-perfect equilibrium. I solved it by value-function iteration.

Findings. With $\beta(1+r) > 1$ and $\varphi > 0$, long-run debt is exactly zero, so the balanced-budget band survives. With $\beta(1+r) = 1$ and $\varphi = 0$, the outcome is Barro ($q = 1$) or Alesina–Tabellini debt ($q < 1$). Some cases cycled. With $\beta(1+r) > 1$ and $q = 1$, the planner saved without limit.

Consequences. (1) The "save until taxes are zero, then stop" planner needs $\beta(1+r) = 1$ *plus* spending shocks, so Versions 4 and 6 switch to that benchmark. (2) Following Battaglini (2026), the cycling may reflect complex equilibrium dynamics rather than numerical error.

6 Version 4: spending shocks with $\beta(1+r) = 1$

Set-up. Version 1 with $\beta(1+r) = 1$. Period-2 spending is $G_2 = \bar{G} \pm \sigma$ with equal probability. Let $\Lambda(G) = \lambda(\tau(G))$ be the MCPF at the tax rate that raises G, and define the buffer premium $P = \mathbb{E}\Lambda(G_2)/\Lambda(\bar{G}) - 1$.

Derivation. At $a = 0$, saving the first rupee gains $[1 - (1-q)\varphi]\,\mathbb{E}\Lambda(G_2)$ and costs $\Lambda(\bar{G})$, so the government saves iff

$$(1-q)\varphi < \frac{P}{1+P} \quad \Longleftrightarrow \quad q > q^* = 1 - \frac{P}{(1+P)\varphi}.$$

Lemma 1 (Λ is convex)**.** *$\lambda' = \varepsilon/(1-(1+\varepsilon)\tau)^2 > 0$ and $\lambda'' = 2\varepsilon(1+\varepsilon)/(1-(1+\varepsilon)\tau)^3 > 0$. Below the peak, $R' > 0$ and $R'' = \varepsilon(1-\tau)^{\varepsilon-2}[(1+\varepsilon)\tau - 2] < 0$. Since $\Lambda' = \lambda'/R'$, we get $\Lambda'' = (\lambda''R' - \lambda'R'')/R'^3 > 0$. Hence $P > 0$.*

Proposition 1 (More risk, more tolerance)**.** *Any mean-preserving spread of G_2 raises $\mathbb{E}\Lambda(G_2)$ (Rothschild–Stiglitz), so it raises P and lowers q^*. The turnover risk a state can tolerate and still save rises with spending risk. (This corrects the sign as I stated it in my 11 October email.)*

Proposition 2 (Raiding is first order, the buffer second order)**.** *$P = \frac{1}{2}\frac{\Lambda''(\bar{G})}{\Lambda(\bar{G})}\sigma^2 + O(\sigma^4)$. Expanding at small tax rates gives $\Lambda(G) = 1 + \varepsilon G + \varepsilon(1+2\varepsilon)G^2 + O(G^3)$. To leading order,*

$$1 - q^* \;\approx\; \frac{\varepsilon(1+2\varepsilon)\,\sigma^2}{\varphi}.$$

The benefit of a buffer scales with the variance of shocks, while the cost of raiding is proportional to φ. This is why small raiding risks dominate, and it also shows that a higher ε means more saving. The numerical curvature at $G \to 0$ is 0.753, 2.012 and 6.054 against the formula's 0.75, 2 and 6.

	Threshold q^* ($\bar{G} = 0.1$, $\varphi = 0.5$)		
ε	$\sigma = 0$	$\sigma = 0.02$	$\sigma = 0.04$
0.25	1.000	1.000	0.998
0.5	1.000	0.998	0.993
1.0	1.000	0.992	0.969

7 Version 5: fungibility of a protected sinking fund

Set-up. Three periods, $\beta(1+r) = 1$. In period 1 the incumbent puts $F \geq 0$ into a fund that can only retire debt. In period 2 (election and shock $\bar{G} \pm \sigma$), a successor cannot touch F but borrows and diverts $x = \varphi \min\{F(1+r), \bar{D}\}$, where $\bar{D}$ is the borrowing headroom; the fund repays this in period 3. With $\bar{D} = \infty$ this is Version 4.

Predictions. (1) No ceiling: protection is useless. (2) $\bar{D} = 0$: planner-level saving. (3) Protection and a ceiling are complements. (4) The objective has a convex kink at $F(1+r) = \bar{D}$, so the choice is all-or-nothing. (5) A ceiling on *net* debt would restore full fungibility.

Numerical check ($\varepsilon = 0.5$, $\bar{G} = 0.1$, $\sigma = 0.04$, $\varphi = 0.5$). The all-or-nothing jump is confirmed: with $\bar{D} = 0.0001$, $F^* = 0.00073$ at $q = 0.995$ and $F^* = 0$ at $q = 0.99$. It holds only when $\bar{D}$ is below the planner's buffer (0.7–5% of $\bar{G}$), because the gain from a buffer is second order while the expected raid is first order in $\bar{D}$.

Mapping to Indian rules. The CSF is held by the RBI, can only redeem liabilities, and withdrawals are capped (terms vary by state and were relaxed in 2020–21). Borrowing is capped by the net borrowing ceiling (about 3% of GSDP a year) and state FRBM Acts, but nothing links the cap to the fund. India ring-fences the fund but does not link borrowing to it. *Open question:* the ceiling limits gross borrowing minus repayments, so redemptions paid from the CSF may free borrowing room rupee for rupee. If so, the withdrawal rule is the effective φ.

8 Version 6 (latest): the long run with raiding

8.1 Environment

- Infinite horizon, $\beta(1+r) = 1$, spending needs G_t i.i.d. and non-degenerate.

- Revenue is $x_t = G_t + (1+r)b_t - b_{t+1}$; if $x_t < 0$ the government pays lump-sum transfers valued one-for-one (as in Battaglini–Coate). So $\lambda \geq 1$, with $\lambda = 1$ at zero taxes.
- Turnover with probability $1-q$, after which assets become $T(b)$.
- *Assumption:* every government maximises citizens' welfare, and raided resources are dissipated.
- Debt is bounded by the natural limit $\bar{b}_N = (R_{\max} - G_H)/r$, or by a ceiling $\bar{D} < \bar{b}_N$ when stated.

Proposition 3 (The equilibrium is a planner problem with a stochastic levy)**.** *The Markov equilibrium solves*

$$V(b,G) = \max_{b'} \tilde{U}\big(G + (1+r)b - b'\big) + \beta\, \mathbb{E}_{G'}\big[q\, V(b', G') + (1-q)\, V(T(b'), G')\big].$$

V is concave and decreasing in b, falls with $1-q$ and with φ, and cannot be raised by a debt ceiling.

Proof outline. An equilibrium value is a fixed point of the Bellman operator, which is a contraction, so it is unique. T is convex, and a concave decreasing function of a convex map is concave, so the operator preserves concavity. Monotonicity in (q, φ) follows because $V(T(b')) \leq V(b')$ and T rises with φ. A ceiling only shrinks a single optimiser's choice set.

Lemma 2 (Euler conditions and the wedge)**.** *Let $\omega_t = \mathbb{E}_t\lambda_{t+1} - \lambda_t$, where λ_{t+1} is next period's realised MCPF. Comparing the gain from borrowing one more rupee today (λ_t) with its expected cost tomorrow:*

Choice of b'	*Optimality*	*Wedge ω_t*
Fund, $b' < 0$	$\lambda_t = \mathbb{E}_t[q\lambda^S + (1-q)(1-\varphi)\lambda^R]$	$(1-q)\varphi\, \mathbb{E}_t\lambda^R \;\geq (1-q)\varphi > 0$
Zero debt	$(1-(1-q)\varphi)\mathbb{E}_t\lambda_{t+1} \leq \lambda_t \leq \mathbb{E}_t\lambda_{t+1}$	≥ 0
Debt, $0 < b' < \bar{D}$	$\lambda_t = \mathbb{E}_t\lambda_{t+1}$	0
At the ceiling	$\lambda_t \geq \mathbb{E}_t\lambda_{t+1}$	≤ 0

The first row is the key one. The government weights the raided outcome by $1-\varphi$ because the money is gone, but that high-cost outcome still happens in full. Without a ceiling, the MCPF is therefore a *submartingale*. For the planner it is a martingale bounded below by 1, so it converges to 1, which is the zero-tax fund.

Theorem 1 (Raiding removes the fiscal anchor)**.** *If $q < 1$, $\varphi > 0$ and there is no ceiling below the natural limit, there is no stationary distribution of (b, G) with $\mathbb{E}\lambda < \infty$. In particular, the planner's zero-tax fund is never a steady state, for any $(1-q)\varphi > 0$.*

Proof outline (by contradiction).

1. *Zero drift.* In a stationary distribution, $\mathbb{E}\lambda_{t+1} = \mathbb{E}\lambda_t$, so $\mathbb{E}\omega_t = 0$. Since $\omega_t \geq 0$, $\omega_t = 0$ almost surely. So no fund is held (there $\omega_t > 0$), and λ is a martingale.
2. *Frozen MCPF.* A martingale bounded below converges almost surely (Doob), so $\lambda_{t+1} - \lambda_t \to 0$. Under stationarity this difference has the same law in every period, so $\lambda_{t+1} = \lambda_t$ almost surely: the tax rate never changes.
3. *Contradiction.* With constant revenue $\bar{x}$, $b_{t+1} = (1+r)b_t + G_t - \bar{x}$, and $\mathrm{Var}(b_t \mid \mathcal{F}_0) = \mathrm{Var}(G)\sum_{s<t}(1+r)^{2s} \to \infty$. (With $\lambda \equiv 1$ and no fund, debt grows by at least G_L each period.) Debt cannot stay bounded.

For the planner, step 3 fails at the zero-tax fund, because transfers vary and absorb the shocks. With raiding, reaching that point requires holding a fund, which is exactly where the MCPF is pushed up. The argument is the martingale method of Chamberlain and Wilson (2000), applied here to raided public funds.

Theorem 2 (The debt ceiling as anchor)**.** *With a ceiling $\bar{D} < \bar{b}_N$:*

(a) a stationary distribution μ exists;

(b) wedge accounting*: $\mathbb{E}_\mu[\,|\omega|\,;\, \text{ceiling binds}] = (1-q)\varphi\,\mathbb{E}_\mu[\lambda^R;\, b' < 0] + \mathbb{E}_\mu[\omega;\, b' = 0]$;*

(c) if $q < 1$ and $\varphi > 0$, the ceiling binds with positive probability; for the planner the probability goes to 0;

(d) $P_\mu(\text{fund held}) \le \mathbb{E}_\mu[\,|\omega|\,;\, \text{ceiling binds}]\,/\,[(1-q)\varphi]$.

Proof outline. (a) The state space is compact and the policy is continuous, so Krylov–Bogolyubov applies. (b) $\mathbb{E}_\mu \omega = 0$; split by the rows of Lemma 2 (the interior contributes 0). (c) If the ceiling never bound, steps 1–3 above would apply on a bounded state space. (d) Follows from (b), using $\lambda^R \ge 1$ and that the zero-debt term is non-negative.

Reading. On average, the ceiling absorbs exactly the upward pressure that raiding and inaction create. Holding a fund a share p of the time requires a ceiling that bites at least $p(1-q)\varphi$ on average, so protection and ceilings are complements (the dynamic version of Version 5).

8.2 Numerical verification

The model is solved by time iteration on the Euler equations with endogenous grid points, which handles the kink at zero debt exactly. Parameters: $\varepsilon = 0.5$, $r = 5\%$, $G \in \{0.06, 0.14\}$. The natural limit is 4.90, and the zero-tax fund is 2.80 (28 years of average spending).

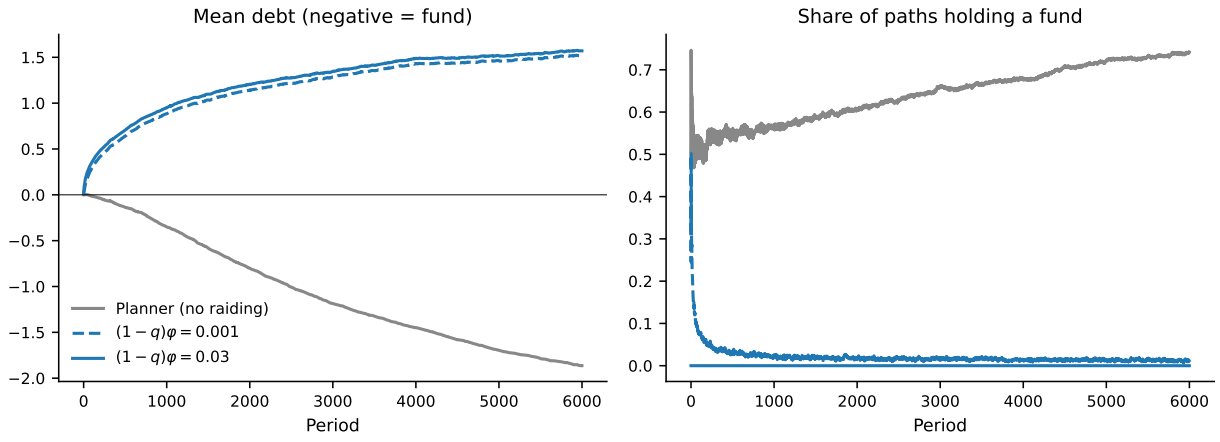


Figure 1: No ceiling, 3,000 paths from zero debt. The planner moves toward the zero-tax fund. With raiding, funds disappear and mean debt rises. An expected loss of 0.1% a year behaves almost like 3%.

Case (with ceiling $\bar{D}$)	Binds	Zero debt	Fund	Raid + inaction wedge	Ceiling wedge	Bound (d)
$q = 0.9,\ \varphi = 0.3,\ \bar{D} = 1.0$	0.28%	0.47%	0	0.000044	0.000044	$0 \leq 0.0015$
$q = 0.9,\ \varphi = 0.3,\ \bar{D} = 0.5$	0.86%	1.21%	0	0.000113	0.000113	$0 \leq 0.0038$
$q = 0.9,\ \varphi = 0.3,\ \bar{D} = 0.25$	2.75%	3.54%	0	0.000343	0.000342	$0 \leq 0.0114$
$q = 0.99,\ \varphi = 0.1,\ \bar{D} = 0.5$	0.70%	0	8.75%	0.000091	0.000092	$0.0875 \leq 0.0917$
Planner, $\bar{D} = 0.5$	$\to 0$	0	95%	0	≈ 0	–

The distributions are stationary (averages over periods 3,000–6,000), and the identity holds to six decimals.

8.3 Limits

1. The $\beta(1 + r) = 1$ benchmark: with $\beta(1 + r) < 1$ the planner does not build a fund either.
2. Dissipated raids: if parties valued their own raids, strategic terms would appear and Proposition 3 would no longer hold.
3. q and φ are reduced forms.
4. Theorem 1 gives no speed of drift (in simulations debt spreads slowly, like a reflected random walk).

9 Questions for discussion

1. Is the $\beta(1 + r) = 1$ benchmark with dissipated raids an acceptable main setting, keeping Versions 1–2 as simple illustrations?
2. Is applying the Chamberlain–Wilson argument to raided public funds a contribution worth building the paper around?
3. Next step: add Alesina–Tabellini disagreement to Version 6 (a debt-region wedge, which should make the ceiling bind more often), or make q endogenous via probabilistic voting (Lindbeck and Weibull 1987; Battaglini 2014)?
4. Is the prediction that borrowing ceilings must bind regularly worth taking to Indian state data (utilisation of the net borrowing ceiling, CSF balances)?

References

Aiyagari, R., A. Marcet, T. Sargent and J. Seppälä (2002), *Journal of Political Economy.*

Alesina, A. and G. Tabellini (1990), *Review of Economic Studies.*

Battaglini, M. (2014), "A dynamic theory of electoral competition", *Theoretical Economics* 9: 515–554.

Battaglini, M. (2026), "Chaos and unpredictability with time inconsistent policy makers", *Theoretical Economics* 21: 241–280.

Battaglini, M. and S. Coate (2008), *American Economic Review.*

Chamberlain, G. and C. Wilson (2000), *Review of Economic Dynamics.*

Khemani, S. (2004), *Journal of Development Economics.*

Lindbeck, A. and J. Weibull (1987), *Public Choice*.

Rothschild, M. and J. Stiglitz (1970), *Journal of Economic Theory*.

Note: derivations were checked, and the numerical code written, with the help of an AI assistant (Claude, Anthropic). Code and full proofs are available on request.